

London Restaurant Data Collection & Matching Methodology

2025-11-29

Overview

This document summarizes the two-step process used to collect and match restaurant data in London, combining Google Places information with Food Hygiene Rating Scheme (FHRS) data.

Step 1: Google Places Data Collection

Objective

Collect comprehensive restaurant data across Greater London using the Google Places API.

Methodology

Grid-Based Sampling Approach:

- Defined a bounding box covering Greater London:
 - Latitude: 51.28° to 51.70°
 - Longitude: -0.51° to 0.33°
- Created a 10×10 grid (100 points) across this area
- At each grid point, performed a Nearby Search with:
 - Radius: 2,000 meters
 - Type: “restaurant”
 - Max results per point: 20

Data Collected:

For each restaurant, the following fields were retrieved:

- Place ID (unique identifier)
- Name and address
- Geographic coordinates (latitude/longitude)
- Rating and number of reviews
- Food types/categories
- Price level
- Opening hours

Deduplication:

- Restaurants were deduplicated by Place ID across all grid points
- Target: 10,000 unique restaurants
- Output: london_restaurants_grid_v2.csv

Key Parameters

```
GRID_ROWS = 10  
GRID_COLS = 10  
RADIUS_METERS = 2000  
TARGET_PLACES = 10000
```

Step 2: Matching with FHRS Health Ratings

Objective

Match Google Places data with official Food Hygiene Rating Scheme (FHRS) records to combine online reviews with government health inspection ratings.

Matching Algorithm

The matching process uses a multi-factor scoring system:

1. Geographic Proximity

- Candidates filtered to within 500 meters of FHRS location
- Distance scoring:

- 50m: score = 1.0
- 150m: score = 0.7
- 300m: score = 0.4
- 500m: score = 0.2

2. Name Similarity

- Fuzzy string matching using token sort ratio
- Minimum threshold: 70% similarity
- Prevents false matches (e.g., different restaurants in same food court)

3. Postcode Validation

- Checks if FHRS postcode appears in Google address
- Additional signal to confirm correct match

Combined Scoring Formula:

`match_score = 0.85 × name_score + 0.10 × distance_score + 0.05 × postcode_score`

- Name heavily weighted (85%) to prevent mismatches
- Minimum combined score threshold: 0.8

Matching Process

Step 1: Candidate Selection

- For each FHRS record, narrow candidates using geographic window:
 - Latitude window: $\pm 0.01^\circ$ (~1.1 km)
 - Longitude window: $\pm 0.015^\circ$ (~1 km)

Step 2: Score Calculation

- Compute match score for each candidate
- Filter out candidates with:
 - Distance > 500 meters
 - Name similarity < 70%
 - Combined score < 0.8

Step 3: Best Match Selection

- Select highest-scoring candidate for each FHRS record
- If no candidates meet thresholds, mark as unmatched

Output Fields

The final matched dataset includes:

From FHRS:

- Business name, address, postcode
- FHRS ID and rating value
- Geographic coordinates

From Google Places:

- Place ID
- Google name and address
- Star rating (1-5)
- Number of reviews
- Food types/categories
- Price level

Match Quality Metrics:

- Match score (0-1)
- Name similarity score (0-100)
- Distance in meters

Quality Checks

The script includes validation to detect:

- Duplicate FHRS IDs in input data
 - Multiple FHRS establishments matching the same Google Place
 - Overall match rate statistics
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Data Integration Results

The final dataset (`london_match_v3.csv`) provides:

1. **Comprehensive Coverage:** Combines official health ratings with user-generated reviews
2. **Geographic Precision:** Location data from both sources
3. **Quality Metrics:** Transparent scoring showing match confidence
4. **Rich Attributes:** Restaurant characteristics from both FHRS and Google

This integrated dataset enables analysis of relationships between official health ratings and public perception (Google ratings/reviews).